The Stock Market and the SpaceX IPO

Week 4

FINS5545: Financial Market Data Literacy

Run It Yourself

This lecture follows three Python scripts you can run line by line in PyCharm.

- `01_spacex_prices_returns.py` — download, clean, plot, and compute returns
- `02_spacex_data_narrative.py` — growth of $1, volatility, the Sharpe ratio, heavy tails
- `03_spacex_vs_benchmarks.py` — compare SpaceX with Tesla, Nvidia, and the Nasdaq-100
- We learn the basics of the stock market through a single market event, the largest IPO in history.

Every figure on these slides is produced by one of these scripts, using live data from Yahoo Finance.

The SpaceX IPO

What Happened

On 11 June 2026, SpaceX sold shares to the public for the first time, in the largest **IPO** in history (SpaceX, 2026, and CNBC, 2026).

- It sold about 556 million shares at an **offer price** of **$135** each, raising roughly $75 billion.
- Trading began on 12 June on the **Nasdaq** stock exchange under the ticker **SPCX**.
- The shares opened at $150, reached $176 during the day, and closed at **$160.95** — up **19%** on the first day.
- This valued the company at about $2.1 trillion and made Elon Musk the first person worth more than $1 trillion.
- A $135 share was worth about $161 by the end of the first day. Where did that jump come from, and what happened next?

Why It Is Controversial (Control)

Many investors and governance scholars criticised the way the company is run (Bebchuk and Kastiel, 2026).

- The shares sold to the public (**Class A**) carry one vote each. Musk and insiders hold **Class B** shares with ten votes each.
- Musk owns about 42% of the company but controls roughly 80–85% of the votes, so public shareholders have little say.
- As a **controlled company**, SpaceX does not need a majority-independent board, and shareholder disputes must go to private arbitration rather than open court.
- Owning a share gives you a claim on the profits, but here it gives you almost no control over the company.

Why It Is Controversial (Price and Regulators)

Critics also questioned the price and asked regulators to step in (India Today, 2026).

- At $135, the company was valued at about 95 times its 2025 revenue, even though it reported a loss for the year.
- The research firm Morningstar called the stock significantly overvalued, with a fair value of about $780 billion — less than half the IPO value.
- One large pension fund refused to invest, calling the governance "catastrophic." US Senator Elizabeth Warren asked the regulator to delay the listing.
- A high price is a forecast about the future. Whether SpaceX is worth $2 trillion depends on profits it has not yet earned.

What Is an IPO?

From Private to Public

An **initial public offering (IPO)** is the first time a company sells its shares to the general public.

- Before the IPO, a company is **private**, so only founders, employees, and selected investors own shares.
- In the IPO, the company sells new shares to raise money. This first sale is the **primary market**.
- After the IPO, those shares trade between investors on a stock exchange. This later trading is the **secondary market**, and the company does not receive that money.
- Companies list to raise money and let early investors sell. Investors buy for a share of future profits.

The Offer Price and the IPO Pop

The **offer price** is set by the company and its banks the night before trading. The **first traded price** is set the next day by buyers and sellers on the exchange. They are rarely equal. The jump between them is the **IPO pop**.

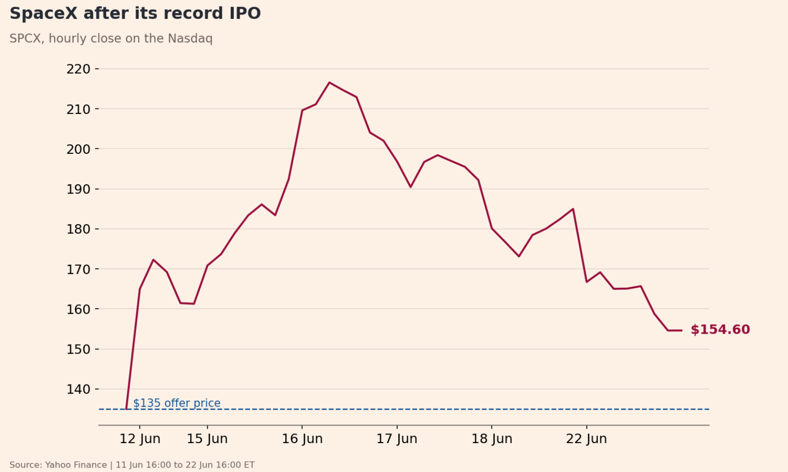
$$\text{pop} = \frac{P_{\text{open}} - P_{\text{offer}}}{P_{\text{offer}}}.$$

For SpaceX, the shares were priced at $P_{\text{offer}} = \$135$ and opened at $P_{\text{open}} = \$150$, giving

$$\text{pop} = \frac{150 - 135}{135} = 0.111 = \mathbf{11\%}.$$

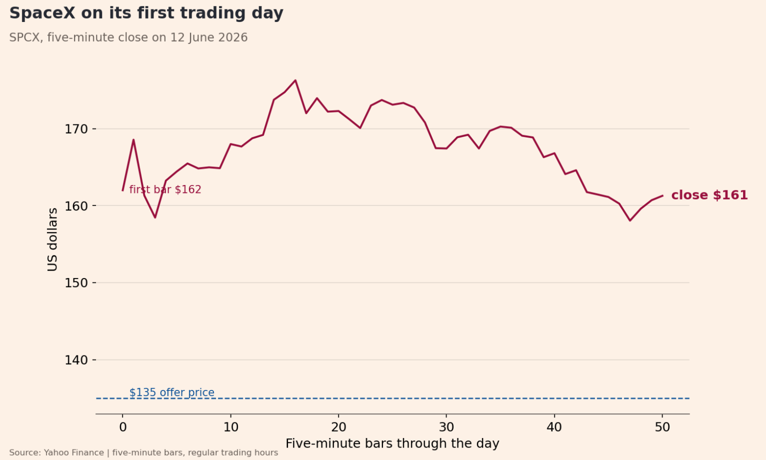
- A large pop means the company sold its shares too cheaply, so money it could have raised instead went to the investors who were allocated shares. Commentators called this "leaving money on the table."
- The offer price is negotiated, while the market price is discovered in trading. The gap is the pop.

The Pop, and the Giveback, in the Data



The price jumps off the $135 offer line (dashed) to about $165 within the first hour, rises to $217 by 16–17 June, then gives back those gains to close at $154.60 on 22 June. Hourly close, New York time.

The First Trading Day, Minute by Minute



Five-minute prices on 12 June. Regular trading did not begin until late morning. The exchange first ran an auction to find a single fair opening price, then continuous trading began.

What Is a Stock Exchange?

How Shares Are Traded

A **stock exchange**, such as the Nasdaq, is a marketplace that matches buyers and sellers of shares.

- Investors send orders through a **broker** (an app or a firm licensed to trade on the exchange).
- A **buy order** says how many shares you want and the most you will pay. A **sell order** says how many you offer and the least you will accept.
- The exchange keeps an **order book**, which lists all the buy and sell orders waiting, sorted by price. A trade happens when a buy price meets a sell price.

Bid, Ask, and the Opening Auction

At any moment the order book has a best **bid** (the highest price a buyer will pay) and a best **ask** (the lowest price a seller will accept).

$$\text{spread} = P_{\text{ask}} - P_{\text{bid}}, \qquad P_{\text{mid}} = \tfrac{1}{2}\left(P_{\text{bid}} + P_{\text{ask}}\right).$$

- The **spread** is the gap between buying and selling. A small spread means the stock is easy to trade.
- On its first day, an IPO does not open at 9:30 with the rest of the market. The exchange collects orders and runs an **opening auction** to find one price that matches the most buyers and sellers. That is why SpaceX's first trade printed late in the morning, near $150.
- The bid-ask spread is the cost of trading immediately. The opening auction sets one fair starting price from many orders.

Measuring Performance with Data

Step 1. Download the Prices

We download SpaceX's price history from Yahoo Finance. We ask for **hourly bars**, one row per hour of trading, each holding the price at the end of that hour (the **close**).

```
prices = download_yahoo_intraday("SPCX", interval="1h", range_="1mo")
```

- This returns about 40 hourly prices, over the six trading days from 12 to 22 June. (19 June was a US public holiday, so the market was closed.)
- We add the $135 offer price as the starting point, so the first return measures the IPO pop.
- The same few lines download any listed stock — only the ticker changes.

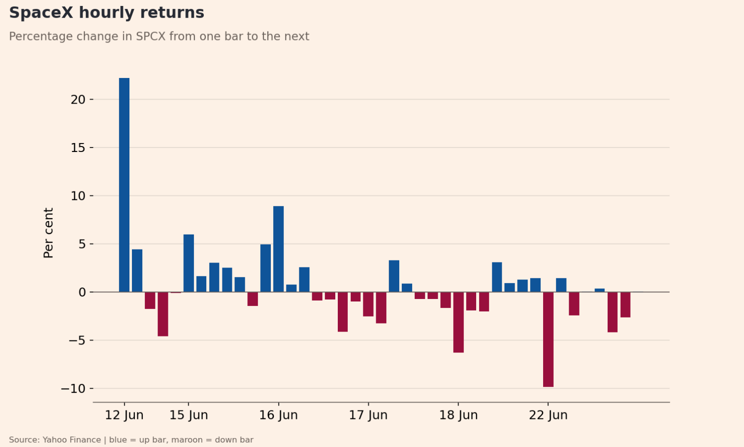
Step 2. Simple Returns

A price level is hard to compare across stocks. A **return** is easier, because it is the percentage change in price from one bar to the next. With price P_t at bar t,

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1.$$

A return of 0.02 means the price rose 2% over that hour. The first return measures the move from the \$135 offer to the first traded price — the pop — and is by far the largest.

Returns, Hour by Hour



Each bar is one hourly return. The tall first bar (**+22%**) is the jump from the $135 offer to the first hourly price. The cluster of maroon bars on the right is the 22 June selloff.

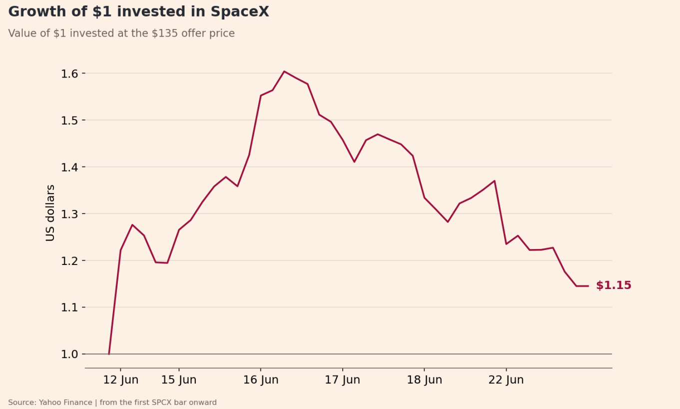
Step 3. Cumulative Return — the Growth of $1

To see the total effect of many returns, we ask what $1 invested at the start would be worth now. We multiply the growth factors $(1 + r_t)$ together.

$$G_T = \prod_{t=1}^{T}(1 + r_t) = \frac{P_T}{P_0}, \qquad \text{total return} = G_T - 1.$$

- Multiplying $(1 + r_t)$ across all bars is the same as dividing the last price by the first price.
- We plot the dollar value, which is always positive, so $1.20 means a 20% gain so far.

The Growth of $1



$1 invested at the $135 offer grew to **$1.15** (a **+14.5%** total return) by 22 June, after peaking near $1.60. An investor who instead bought at the first public trade, near $165, was **down about 6%** over the same window — the early pop had faded.

Step 4. Volatility — Measuring Risk

Volatility measures how much the returns bounce around. It is the standard deviation of the returns. With n returns and mean $\bar{r}$,

$$\bar{r} = \frac{1}{n}\sum_{t=1}^{n} r_t, \qquad \sigma = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}(r_t - \bar{r})^2}.$$

A larger σ means a riskier, more unpredictable stock. To compare across different bar sizes we **annualize**. If there are N return periods in a year, volatility scales with the square root of time,

$$\sigma_{\mathbf{ann}} = \sigma\sqrt{N}.$$

- Volatility is the standard deviation of returns, and higher volatility means higher risk.

Step 5. The Sharpe Ratio — Reward per Unit of Risk

A high return is only impressive if the risk taken was not too high. The **Sharpe ratio** divides the return earned above the safe **risk-free rate** r_f by the volatility (Sharpe, 1966).

$$\text{Sharpe} = \frac{\bar{r} - r_f}{\sigma}.$$

Annualizing the top and bottom (mean by N, volatility by $\sqrt{N}$) gives

$$\text{Sharpe}_{\text{ann}} = \frac{(\bar{r} - r_f)\,N}{\sigma\sqrt{N}} = \frac{\bar{r} - r_f}{\sigma}\,\sqrt{N}.$$

- A higher Sharpe ratio means more reward for each unit of risk, while a negative Sharpe ratio means the stock lost money. Over this short window we set $r_f = 0$.

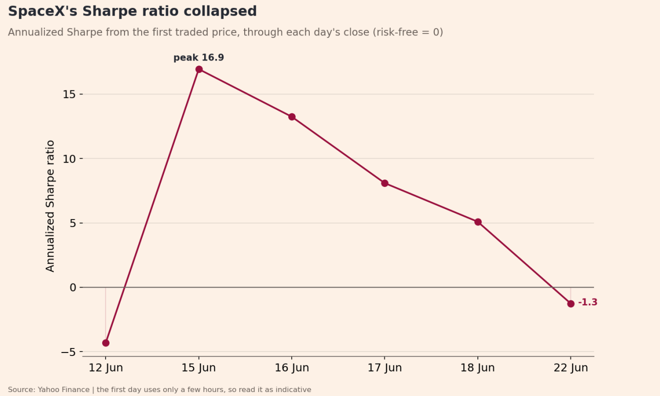
The SpaceX Scorecard

Putting the pieces together for the hourly data, measured from the $135 offer price.

Measure	Value
First price (offer)	$135.00
Last price (22 Jun)	$154.60
Total return	+14.5%
Average hourly return	0.44%
Annualized volatility	201%
Annualized Sharpe ratio	3.8

- Treat the annualized figures with caution. A volatility of 201% and a Sharpe near 4 come from only six days of a brand-new stock. We return to this on the last slide.

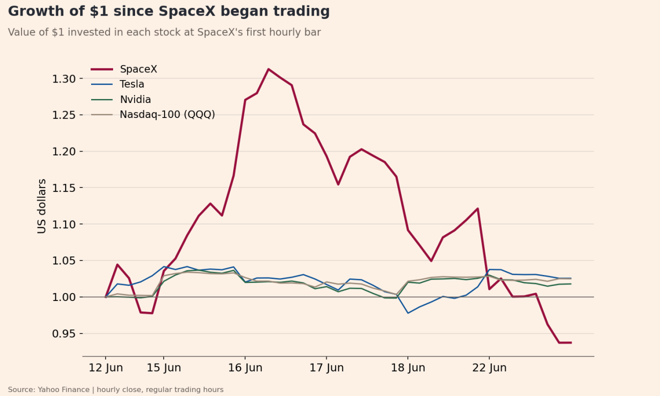
The Sharpe Ratio Collapsed Over the Week



The single scorecard number hides the story. Measured from the first traded price and recomputed at the end of each day, the annualized Sharpe ratio rocketed to **+17** during the rally, then fell every day to −1.3 after the 22 June selloff. The first day uses only a few hours, so read it as indicative.

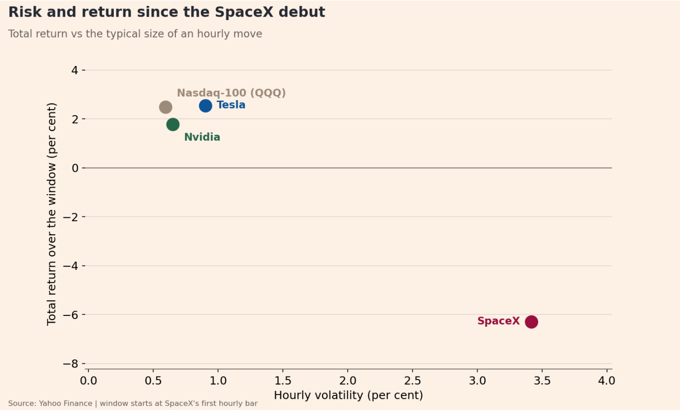
Was SpaceX Special? Compare to Other Stocks

Growth of \$1 — SpaceX vs Other Stocks



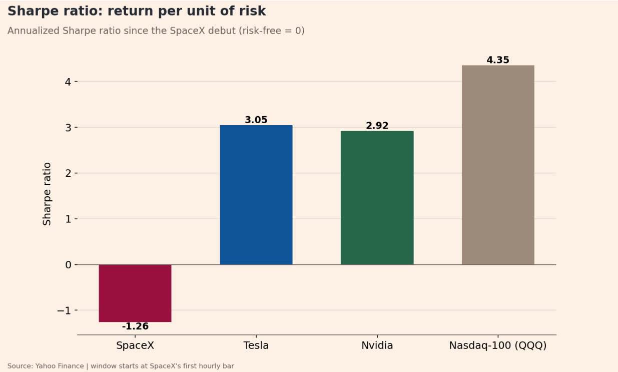
\$1 in each stock from SpaceX's first hour. To compare fairly, SpaceX starts at its first traded price, not the offer (the others have no offer price). SpaceX raced ahead, then gave it all back, so by 22 June it was **down about 6%**, while Tesla, Nvidia, and the Nasdaq-100 were each up 2 to 3%.

Risk and Return Together



Each dot is one stock, with risk (the typical size of an hourly move) across and return up. SpaceX swung by far the most and was the only one to end the window with a loss. The highest risk produced the worst return here, not the best.

More Risk Did Not Mean More Reward



Sharpe ratio (return per unit of risk). SpaceX's is **negative** (−1.26), because it lost money over the window, so each unit of risk was punished. The diversified Nasdaq-100 had the best Sharpe ratio (**4.35**), with a small return and small risk.

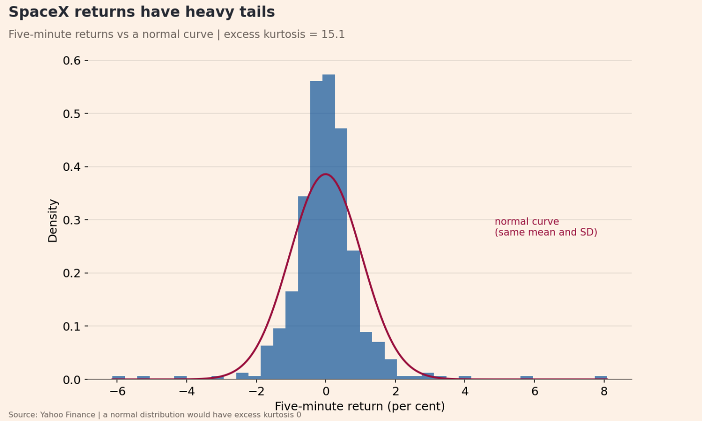
The exciting stock was the worst risk-adjusted bet, while the boring, diversified market was the best.

A Fact About Returns — Heavy Tails

Returns Are Not Normally Distributed

- If we collect many returns and draw their distribution, it does not match the familiar bell curve (the **normal distribution**).
- Observed returns have a taller peak and **heavy tails**. Extreme moves happen far more often than a normal curve predicts. This shape is called **leptokurtosis**.
- It is one of the most reliable facts about asset returns, across markets and time horizons (Cont, 2001).
- We measure it with **excess kurtosis**. A normal distribution scores 0, and a positive number means heavier tails.
- We need many observations to see the shape, so we use the five-minute returns rather than the few dozen hourly ones.

SpaceX Returns Have Heavy Tails



Five-minute returns (blue) against a normal curve with the same mean and standard deviation (maroon). The peak is taller and the tails are heavier than the bell curve. The excess kurtosis is 15.1, far above the 0 of a normal distribution.

Read These Numbers with Care

This is a fun example, but it is a small and unusual sample. Keep three cautions in mind.

- **Six days is not a year.** Annualizing returns and volatility from one week of data gives extreme figures (a 201% volatility, a Sharpe near 4) that will not last. The longer the history, the more trustworthy the annual numbers.
- **A brand-new IPO is special.** The first days of trading are unusually volatile as the market discovers a price. They do not represent the stock's normal behaviour.
- **In-sample only.** We measured the past. None of this tells us what SpaceX will do next.
- The methods are the lesson, not the numbers. The same code on years of data gives figures you can rely on.

Summary

- An **IPO** is a company's first share sale to the public, and the **pop** is the jump from the offer price to the first traded price.
- A **stock exchange** matches buyers and sellers, and the price is set by orders in the book, not by any single party.
- A **simple return** is the percentage change in price, and the **growth of $1** multiplies the returns together.
- **Volatility** measures risk, and the **Sharpe ratio** measures reward per unit of risk — SpaceX took the most risk and, over this window, earned the worst return.
- Returns have **heavy tails**, so extreme moves are more common than a normal distribution predicts.

Run it yourself. The three scripts in `week4/scratch/spacex_ipo/` reproduce every figure on these slides.

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